

1. Explain uses of ping command in network ?

The Ping Command is mainly used for two System (PC) is connected to data sharing data PC(A) to PC(B) ^{to check} ^{to check} how we know to data is sharing ^{properly} that why we use Ping Command both System is communicating which each other ^{or not} if data shared means that standard called OSI layer.

2. Explain netbios protocol ?

Netbios $\rightarrow$ (network basic Input output System) It is used when two system PC is connected with local area network (LAN) when use to share data ^{using} IP address using netbios we can share with computer user name to share data with both computer in local area network using netbios protocol.

3. Explain SMB protocol ?

SMB $\rightarrow$ Server Message Block that allows computer to share file, folder, printer, network drives. When have two system PC(1) share its data $\rightarrow$ PC(2) using ~~file~~ or file means you know the destination PC's IP address & user name. So PC(1) share data PC(2) ask to SMB Protocol open shared folder it give access means it will open or else it will not give permission (read, write, etc), ^{permission} 50.0.0.10 to 50.0.0.20 ?

4. Explain ping process from 50.0.0.10 to 50.0.0.20 ?

This system is connected physically with this can't share data so that we are assign for PC(A) IP address $\rightarrow$ 50.10 For PC(B) IP address 50.20 using this share the data PC(A) to PC(B) how we know about it is communicating or sharing the data. that why we use ping (cmd) to check both system has a communication if connection is there means data is sharing that standard it called OSI layer that help ^{the standard process of data sharing} ^{Application $\rightarrow$ share data using ICMP} ^{layer}

[layer $\rightarrow$ Presentation : Format the data] [layer $\rightarrow$ Session : Timing at its send data] [layer $\rightarrow$ Transport $\rightarrow$ 3 $\rightarrow$ Network : still using IP Protocol this process called Packet encapsulation] [2 Data Link : dm Ethernet don't know about dmac so we call help (ARP) is find the destination in with help of dIP]. And [ARP request

Source mac: 0:0A

D mac : FF:FF:FF:FF:FF:FF that mean is a broadcast mac ^{send the} ^{physical} ^{physical} Request to All physical

Electrical signal

All the PC will have receive the request because it Broadcast so that All them open then drop because this IP address not our who's IP is that system will send reply using ARP reply $\rightarrow$ And then it will contact has unicast $\rightarrow$ ARP table store dmac dIP $\rightarrow$ And then

share one copy to Ethernet and it store dmac this process name called frame encapsulation.

And then it will move to physical layer to share the ^{binary} ^{tool} ^{to} ^{data} ^{to} ^{destination} like this OSI layer process.

Source ID
Sender mac
Target IP
Target mac